MA2104(3)

Test 2

Wednesday, 8 October 2025

Time: 40 minutes Total marks: 15

1. (6 points) Determine all $z \in \mathbb{C}$ such that

$$\tan z = 2 + i.$$

2. (4 points) Evaluate

$$\int_{\gamma(0,2)} f(z)\,\mathrm{d}z, \quad \text{where} \quad f(z) = \operatorname{Re} z.$$

3. (5 points) Prove that $u(x, y) = e^{-y}(2x\cos x - (2y + 3)\sin x)$ is harmonic.

Total: 15 marks